

Research Statement

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My long-term research goal is to enable robots to assist humans in various daily work intelligently and reliably. To achieve this goal, I research novel robot control approaches and use them to enable skillful and robust robot actions, especially in **adversarial environments with disturbances** and **novel environments unseen** in the training data. The approaches I proposed currently are based on the **modular structure design** of neural networks and the **adversarial pipeline design** of the training process. Robot policies trained with conventional reinforcement learning or supervised learning approaches usually cannot adapt to a different environment or adversarial disturbances. To enable a fast adaptation to novel environments, I developed the Policy Stitching algorithm [1] and the Perception Stitching algorithm [2] that are capable of stitching neural network modules trained in different environments and effectively deploying the reassembled network to a novel environment unseen in previous training. Furthermore, to train the robot policy to be robust against internal and external disturbances, I proposed the Adversarial Skill Learning method [3] that introduces an adversary agent to disturb the protagonist agent during the reinforcement learning process, which consequently trains the protagonist agent to be robust to malicious adversarial attacks as well as various random perturbations.

Through my work on **training neural network modules to be reuseable** in other environments, I have found that the latent representation misalignment issue [4] between the encoders and decoders trained in different environments is the major problem that diminishes the performance of the reassembled policy in a new environment. This observation drives my research into developing techniques inspired by the relative representation algorithm [4] and covariance regularization method [5, 6, 7, 8] to enforce the invariance at the latent space of different encoders. Our experience has shown that training latent states that have independent features and invariant to isometric transformation can significantly alleviate the latent space misalignment and improve the reassembled policy performance. However, my current approaches cannot achieve very satisfying performance on long-horizon dexterous tasks yet, leading to my future research to further understand the latent space of neural networks for standardizing the interfaces between different network modules.

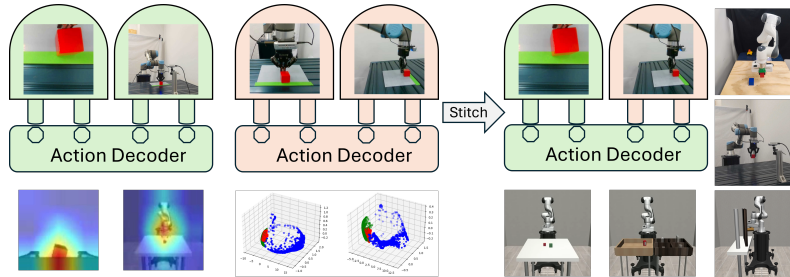


Figure 1: Perception Stitching [2]. “Policy A” was trained with an in-hand camera and a front-view camera. “Policy B” was trained with a close-up camera and a side-view camera. Perception Stitching enables zero-shot stitching of the original Policy A and B by reusing their relevant components for each sensing configuration to form a “Policy C”. “Policy C” can maintain strong zero-shot transfer performance with an in-hand camera and a side-view camera.

In an endeavor to **train robust control policy** for robots, I observed that conventional data augmentation methods that inject random noise into the trajectories data generally cannot be effective against more malicious disturbances beyond random noise. Inspired by the training pipeline of the Generative Adversarial Network (GAN) [9], I proposed the Adversarial Skill Learning method that trains a protagonist agent for accomplishing the task and an adversary agent for obstructing the task

iteratively. After this competitive GAN-style training process, the adversary agent learns dexterous skills to attack the protagonist agent, while the protagonist agent is trained to be robust to various malicious disturbances. While my proposed method shows pronounced advantage to vanilla data augmentation, it still requires manually tuning the optimal amplitude range of the disturbance for the training. In my future research, I would like to empower the Adversarial Skill Learning framework with the recent advances in the diffusion policy and the foundation models. Along this path, I believe I can push the limits of robot control robustness to reliably deploy robots in the real world.

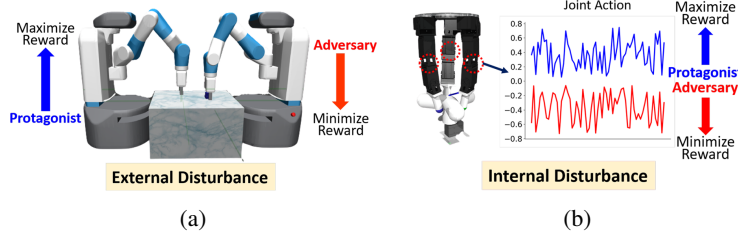


Figure 2: Adversarial skill learning for robust manipulation [3]. The protagonist tries to maximize the reward while the adversary tries to minimize it. (a) An external disturbance, the protagonist is disturbed by another adversary robot. (b) An internal disturbance, the protagonist joint action is under the disturbance of adversarial action on each joint of the robot.

Recently, I began to explore the application of foundation models in my research agenda.

Previous research has already found that large language models (LLMs) can effectively design and improve the reward functions of some robotic tasks trained with reinforcement learning (RL) [10, 11]. We experimented this idea on some difficult quadruped robot locomotion tasks and found that the training process consumes very expensive computation resources for parallel reward functions searching, yet the trained robots sometimes cannot achieve satisfying performance, showing unnatural gait patterns or struggling to accomplish the tasks. In the short term, my research objective is the application of vision-language models (VLMs) on model-based reinforcement learning for more efficient training computation and performance that better satisfies the human preferences [12]. In the long term, I would like to explore leveraging the LLMs and VLMs for state and action abstraction learning, which I believe is a promising path towards general robotic intelligence.

Looking further, my very long-term goal is to develop general embodied intelligence that assists people in every aspect of their work and lives. To achieve this goal, we need to have deeper understanding on the mechanism of both the natural intelligence and the artificial intelligence. Based on the advances in understanding the root source of “intelligence”, we can build robotic systems that push the limits of robustness and adaptation capabilities that I previously focused on, as well as many other performance facets that I am excited to explore.

References

- [1] P. Jian, E. Lee, Z. Bell, M. M. Zavlanos, and B. Chen. Policy stitching: Learning transferable robot policies. *arXiv preprint arXiv:2309.13753*, 2023.
- [2] P. Jian, E. Lee, Z. Bell, M. M. Zavlanos, and B. Chen. Perception stitching: Zero-shot perception encoder transfer for visuomotor robot policies. *arXiv preprint arXiv:2406.19971*, 2024.
- [3] P. Jian, C. Yang, D. Guo, H. Liu, and F. Sun. Adversarial skill learning for robust manipulation. In *2021 IEEE International Conference on Robotics and Automation (ICRA)*, pages 2555–2561. IEEE, 2021.
- [4] L. Moschella, V. Maiorca, M. Fumero, A. Norelli, F. Locatello, and E. Rodolà. Relative representations enable zero-shot latent space communication. *arXiv preprint arXiv:2209.15430*, 2022.
- [5] A. Bardes, J. Ponce, and Y. LeCun. Vicreg: Variance-invariance-covariance regularization for self-supervised learning. *arXiv preprint arXiv:2105.04906*, 2021.

- [6] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny. Barlow twins: Self-supervised learning via redundancy reduction. In *International conference on machine learning*, pages 12310–12320. PMLR, 2021.
- [7] I. Higgins, L. Matthey, A. Pal, C. P. Burgess, X. Glorot, M. M. Botvinick, S. Mohamed, and A. Lerchner. beta-vae: Learning basic visual concepts with a constrained variational framework. *ICLR (Poster)*, 3, 2017.
- [8] I. Higgins, D. Amos, D. Pfau, S. Racaniere, L. Matthey, D. Rezende, and A. Lerchner. Towards a definition of disentangled representations. *arXiv preprint arXiv:1812.02230*, 2018.
- [9] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio. Generative adversarial networks. *Communications of the ACM*, 63(11):139–144, 2020.
- [10] Y. J. Ma, W. Liang, G. Wang, D.-A. Huang, O. Bastani, D. Jayaraman, Y. Zhu, L. Fan, and A. Anandkumar. Eureka: Human-level reward design via coding large language models. *arXiv preprint arXiv:2310.12931*, 2023.
- [11] Y. J. Ma, W. Liang, H.-J. Wang, S. Wang, Y. Zhu, L. Fan, O. Bastani, and D. Jayaraman. Dreureka: Language model guided sim-to-real transfer. *arXiv preprint arXiv:2406.01967*, 2024.
- [12] P. F. Christiano, J. Leike, T. Brown, M. Martic, S. Legg, and D. Amodei. Deep reinforcement learning from human preferences. *Advances in neural information processing systems*, 30, 2017.